Problem Chosen	2021	Team Control Number
C	MCM/ICM Summary Sheet	2021002

Triage Public Hornet Reports under Sparse Verification

Summary

Washington must identify credible Asian giant hornet reports while avoiding the false inference that a quiet reporting stream proves absence. We separate biological spread, report classification and surveillance detection, using only the supplied reports, image mapping and biological attachment.

The 4440 unique reports contain 14 positives, 2069 negatives and 2357 unresolved outcomes. Positivity among adjudicated reports is 0.672% (95% Jeffreys interval 0.386–1.095%); unresolved reports are never assigned negative labels. Ninety reports have missing, very old or inconsistent detection dates, and laboratory-completion times are absent. Image pixels were unavailable, so our classifier uses only report-time text, locations and dates.

We compare prevalence, geographic smoothing, text and combined regularized logistic models. Spatially grouped development selects the combined model before a temporal holdout of 1697 later adjudicated reports. Its top 85 candidates contain all six positives, with precision 7.06% and average precision 0.1142. However, its Brier score of 0.004701 is worse than the prevalence baseline, and its mean probability is 1.595% against observed positivity of 0.354%. Removing explicitly institutional records reduces top-85 recall to three of four positives. The score targets adjudicated-report positivity; unresolved positivity needs a distinct transport assumption. We therefore recommend priority bands, expert verification and a declared 20% exploration allocation, rather than automatic decisions from apparently precise probabilities.

All nine 2020 confirmed locations lie within 29.894 km of a 2019 confirmed sighting. This supports conditional local planning envelopes, but does not identify a propagation rate: occurrence and detection effort are confounded, reports can be delayed, and new introductions remain possible. A retrospective 30-km envelope is not a calibrated prediction region.

New reports should be scored immediately, laboratory outcomes reconciled weekly, and model replacement reviewed monthly or after a confirmed geographically novel event. Timestamped adjudications and recorded exploration sampling are necessary for a credible prospective assessment.

Eradication evidence must include coverage and a lower bound on system detection sensitivity. Under conditionally independent active seasons, zero-detection likelihood under persistence is $(1 - s)^K$: sensitivity 0.8 requires two negative seasons to fall below 0.05, whereas sensitivity 0.3 requires nine. These are conditional scenarios; the supplied public reports do not identify s and do not establish eradication. Our two-page memorandum gives WSDA a practical queue policy and an evidence standard for future surveillance.

Keywords: verification bias; rare-event classification; spatial surveillance; nondetection; report triage

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1 The decision and the limits of the observations

The Washington State Department of Agriculture (WSDA) needs to find a small number of credible *Vespa mandarinia* reports within a much larger stream of look-alike sightings. A report can justify a follow-up without proving that a nest exists. Conversely, an uninformative report cannot establish absence. We organize the analysis around three different objects: the insect's biological occurrence, the probability that an incoming report is positive, and the probability that surveillance would detect a population if it persisted.

The supplied data end on October 23, 2020. We analyze the five requested questions using the original reports, the image-to-report mapping, and the provided biological background [1, 2, 3]. The mapping is used to audit coverage. The image archive could not be acquired, so image content contributes no evidence. This leaves a complete tabular route, but removes a potentially valuable source of morphological confirmation.

Our main decision is a ranked investigation queue. We test it on later adjudicated reports and separately examine what those tests cannot establish. The spread analysis produces conditional spatial scenarios, while the eradication analysis states the surveillance conditions needed for a meaningful negative result. Neither is obtained by treating the report count as a hornet population count.

2 Data contract, audit and assumptions

2.1 What a row and a label mean

The workbook contains 4440 unique report identifiers. Positive ID is a laboratory-confirmed Asian giant hornet; Negative ID excludes that identification. Unverified means the information was insufficient to decide, and Unprocessed means no classification had yet been made. Table 1 preserves these distinctions. Only the 2083 adjudicated reports provide binary reference outcomes. We call the remaining 2357 reports unresolved; that does not imply that WSDA should repeat an investigation of every historical unverified record.

Table 1. Classification states in the original reports

State	Reports	Percent	Statistical role
Positive ID	14	0.32	Reference positive
Negative ID	2069	46.60	Reference negative
Unverified	2342	52.75	Unknown outcome
Unprocessed	15	0.34	Unknown outcome

Among adjudicated records, positivity is 0.672%, with a 95% Jeffreys interval of 0.386–1.095%. This is a conditional report proportion, not an estimate of insect prevalence in Washington. With no assumptions about unresolved outcomes, the fraction of all reports that could be positive lies between 14/4440=0.315% and 2371/4440=53.401%. This intentionally broad identification bound shows how much the verification mechanism matters.

The image mapping has 3305 entries associated with 2127 unique report IDs. Every

mapped ID exists in the report table; joining individual attachments would multiply some reports. We retain one row per GlobalID. Filenames, media types and attachment counts are excluded from the predictor because neither their receipt-time availability nor their relation to laboratory follow-up is established.

2.2 Dates, dependence and information availability

Submission dates run from January 15 to October 23, 2020. Detection dates include six missing values, 66 dates before 2019, and 18 dates after their own submission date. These 90 records receive an invalid-date flag for the temporal features. Their raw values are preserved. The 2019 cutoff defines the recent-outbreak feature window; it is not a claim that every older sighting is false. Detection month and lag are set to neutral values for flagged records, and the flag itself remains available to the model.

For valid dates, the median submission lag is 0 days and the 90 th percentile is 8 days. The 14 confirmed records have a median lag of 10.5 days. A 2019 observation can therefore arrive many months into 2020. Figure 1 displays submission volume; it measures the incoming workload, not biological growth.

Notes are blank in 537 records. More seriously, some positive notes describe an already destroyed colony or an institutionally captured specimen. A predeclared narrow flag identifies four notes with explicit registry-style phrases: “WSDA submitted,” “captured by WSDA,” “reported to provincial government,” or “colony that was found and destroyed.” All four are positive. Removing these records gives a sensitivity analysis of ordinary reports; it is not a complete public-versus-staff census.

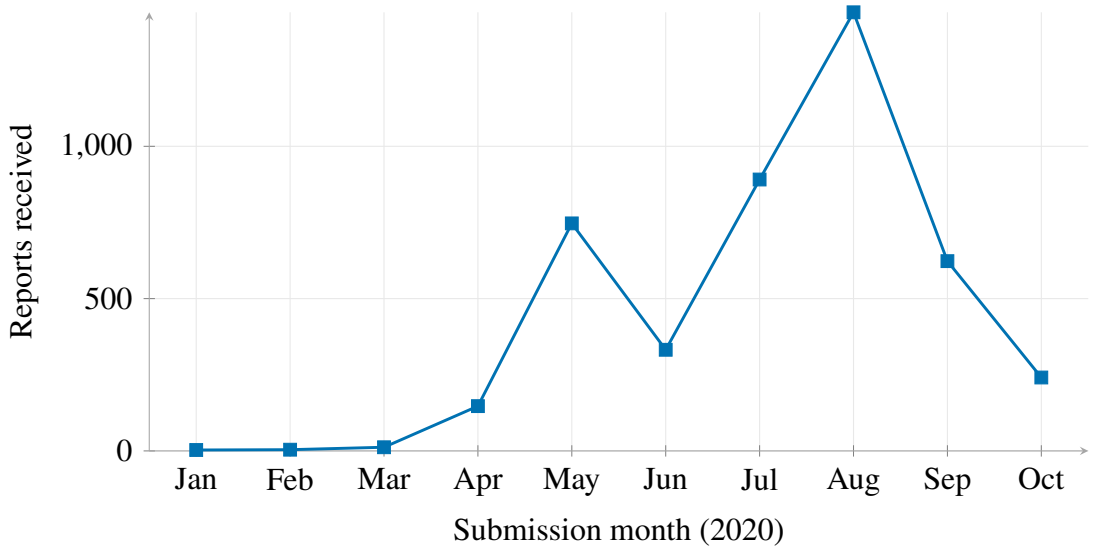


Figure 1. Report arrivals. Counts are grouped by submission month; October is incomplete through the 23rd. These are reporting workloads, not population counts.

Laboratory comments and status are post-investigation information. They are excluded from prediction. GlobalID is retained only for identity and evidence linkage. Coordinates, Notes, reported detection date and receipt date are used as the information available at receipt. The official coordinate conversion may itself have occurred after receipt; the supplied table has

no timestamp for that conversion. Operational use requires locating the submitted address before scoring.

The workbook also lacks laboratory-completion dates. We can prevent training on reports submitted after an evaluation cutoff, but cannot verify that every earlier label was already known then. Our forward test is therefore an optimistic retrospective proxy for deployment. It must be repeated prospectively with actual label-availability timestamps before performance guarantees are made.

2.3 Assumptions, reasons and consequences

We state assumptions where they affect a conclusion, rather than treating them as conveniences that disappear after fitting.

1. **Outcome reference.** Adjudicated Lab Status defines the training target. This follows the supplied data dictionary. Incorrect adjudications would affect both fitting and evaluation; no independent specimen review is available.

2. **Geographic interpretation.** Latitude and longitude are geographic degrees. Their ranges and the original problem map are consistent with this interpretation. The datum and address-location error are unknown; kilometer-scale distances are useful, while exact boundary membership and a 106-metre distance margin are not secure.

3. **Conditional transport.** A fitted probability applies first to reports selected for adjudication. Applying it to unresolved reports assumes a sufficiently similar relation between features, selection and positivity. We do not assume that relation is true; declared verified-to-unresolved odds scenarios show its effect.

4. **Dependence.** Related reports can refer to one local episode. Spatial blocks are used in development and resampling. A 2-km proximity graph collapses 14 positive observations to eight components, including components of three and five reports. These are proxies for dependence, not established nest identities.

5. **Seasonal geometry.** The given 30-km queen nest-establishment estimate is a scenario scale. Annual reproduction supports a generational interpretation. It is not a deterministic upper bound, and independent introductions remain possible.

6. **Eradication sensitivity.** A numerical absence claim requires a lower bound on surveillance detection probability conditional on persistence. The supplied reports do not identify that probability; any numerical eradication calculation is explicitly conditional.

3 Can the biological spread be predicted?

3.1 An identifiable pattern and an unidentifiable process

The confirmed observations are spatially concentrated near the international border, with a separate 2019 Vancouver Island record. They identify where confirmed reports occurred. They

do not identify the number of colonies, the distribution of queens, or the surveillance effort outside those locations.

To see the limitation formally, let Z_{jt} denote presence in cell j during season t , let $\psi_{jt} = P(Z_{jt} = 1)$, and let q_{jt} be the probability that presence generates a verified report. In a simple observation model,

$$P(Y_{jt} = 1) = \psi_{jt} q_{jt}. \tag{1}$$

One observed probability c is consistent with every $\psi \in [c, 1]$ and $q = c/\psi$. More reports cannot by themselves distinguish greater spread from greater detection or public attention. In a count formulation, expected verified reports likewise multiply latent occurrence by reporting and verification effort. The missing denominators and repeated site visits prevent estimating these factors separately.

A diffusion model or logistic growth curve would therefore produce precise-looking parameters for an unidentified biological process. We do not fit either. The background describes an annual life cycle with overwintering queens and autumn production of reproductives [3]; it supports seasonal planning, but not a measured local growth rate. A decline in winter reports is compatible with persistent overwintering queens.

3.2 A conditional spatial forecast and its check

Let A_t be the set of known confirmed anchor locations at the start of a planning period. An annual establishment scenario with scale r has an envelope

$$\mathcal{E}_{t+g}(r) = \bigcup_{a \in A_t} \{x : d(x, a) \le gr\}, \qquad r = 30\,\text{km}, \tag{2}$$

where g is the number of reproductive cycles and d is a great-circle distance. This is a surveillance planning region conditional on local establishment without a new introduction. It is neither a probability contour nor proof that locations outside it are safe.

We compute distances on a sphere of mean radius 6371.0088 km using

$$d = 2R \arcsin \sqrt{\sin^2 \frac{\Delta\phi}{2} + \cos\phi_1 \cos\phi_2 \sin^2 \frac{\Delta\lambda}{2}}. \tag{3}$$

An independent calculation uses the spherical dot-product formula. Both agree to within 1 millimetre of the reported distances under this same spherical model; that numerical agreement does not reduce address-location uncertainty.

The confirmed sample includes Canadian locations; it is a cross-border observation set, not a Washington-only census. Using all five 2019 confirmed sighting locations as anchors, all nine 2020 confirmed locations fall inside a 30-km nearest-anchor envelope. Eight are within 10 km; the largest nearest-anchor distance is 29.894 km. Figure 2 gives all nine distances rather than smoothing them into a front. The 30-km margin of approximately 0.106 km is smaller than a plausible unrecorded address-location error; we cannot claim a sharp biological boundary.

This retrospective check uses 2019 records that were not all available early in 2020. At a January 31 receipt cutoff, only two confirmed anchors were in the table. Eleven of the twelve subsequently submitted positive records are within 30 km of those anchors. The exception lies

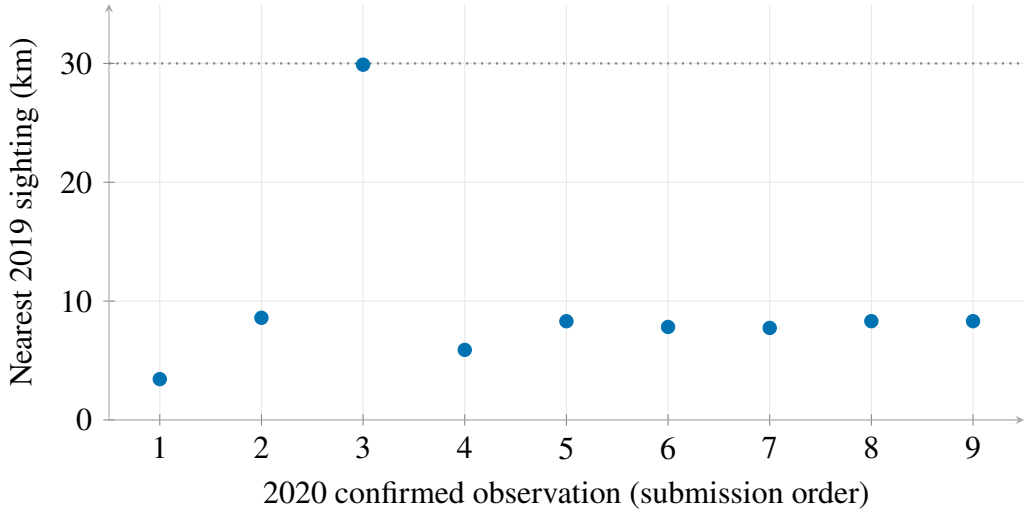


Figure 2. Nearest-anchor distances. Each point is one confirmed 2020 report; the horizontal reference is the declared 30-km scenario. Report identity and exact distances are preserved in the computational outputs.

92.633 km away and describes the Vancouver Island colony detected in September 2019 but submitted in February 2020. It is delayed information about a separate earlier location, not evidence of 92 km of spread after January. This example shows why an animation sorted only by submission date gives a misleading spread history.

Observation locations also need not be nest locations. The provided background describes worker foraging mostly within 1–2 km and up to 8 km [3]. Under a declared scenario allowing 8 km from a parent nest to its observed worker, 30 km for queen establishment, and 8 km from a descendant nest to its observed worker, an observation-to-observation envelope could be 46 km rather than 30 km. Neither radius is a calibrated 95% prediction bound. We recommend 30-, 46- and 60-km planning envelopes as sensitivity scales, with the 60-km scale representing a deliberately wider operational scenario rather than a newly estimated dispersal parameter.

The 2019-to 2020 centroid shift is 28.235 km. It combines a changing mixture of island and mainland reports, delayed submissions and repeated local observations. It is not a spread velocity. These data support local concentration and broad annual risk scenarios, but cannot supply a defensible point forecast of the next nest, a daily propagation rate, or a statewide probability of establishment.

4 A report-time model of mistaken identification

4.1 Target, candidates and fixed feature design

For a report i , define $Y_i = 1$ for Positive ID and $Y_i = 0$ for Negative ID. The first target is

$$p_V(x) = P(Y = 1 \mid X = x, V = 1), \qquad m_V(x) = 1 - p_V(x), \tag{4}$$

where $V = 1$ means the report was adjudicated. The complement m_V is the model's likelihood of mistaken species identification in that conditional population. Unverified and Unprocessed are not coded as $Y = 0$.

We compare four fixed candidates. The prevalence baseline predicts a Jeffreys-smoothed training fraction $(k + 1/2)/(n + 1)$. A transparent geographic smoother weights adjudicated reports by distance:

$$\widehat{p}_G(x) = \frac{20\widehat{p}_0 + \sum_{i\in\text{train}} K_i(x)Y_i}{20 + \sum_{i\in\text{train}} K_i(x)}, \quad K_i(x) = \exp[-d(x, x_i)^2/(2h^2)], \quad h = 30\,\text{km}. \tag{5}$$

The 20-report pseudocount limits extreme predictions in sparsely observed locations. It is a declared regularization choice, not an estimated biological parameter. Bandwidths 10 and 60 km are examined as sensitivity cases, not selected using the later labels.

The text candidate uses lowercased unigram and bigram TF–IDF features, sublinear term frequency, a minimum training-document frequency of 2, and at most 2500 features [5]. The combined candidate adds standardized latitude, longitude, detection-month sine and cosine, $\log(1 + \text{lag})$, the invalid-date flag and the blank-note flag. Both fit regularized logistic regression [4]:

$$\widehat{p}(x) = \frac{1}{1 + \exp[-(\widehat{\beta}_0 + x^\top\widehat{\beta})]}, \qquad \min_{\beta_0,\beta} \sum_i \{-Y_i \log p_i - (1 - Y_i)\log(1 - p_i)\} + \frac{1}{2}\|\beta\|_2^2. \tag{6}$$

The implementation uses $C = 1$, no class reweighting, and an unpenalized intercept. Penalization controls sparse positive data and a large vocabulary. We do not balance classes artificially: that would change the probability target unless corrected.

Every vocabulary, inverse-document frequency and standardization is learned only from each training fold. Words are predictive proxies for what a reporter wrote, not morphological observations. Negation, uncertain size descriptions and institutional phrases can all defeat a bag-of-words interpretation. We therefore assess text as a baseline and report its contribution through withheld performance, without assigning invented biological labels to the prose.

4.2 Development and a protected temporal evaluation

The fixed information cutoff is June 30, 2020. The development set contains 386 adjudicated reports and eight positives. A fixed 10-km grid provides five spatially grouped folds; an entire grid cell stays in either training or validation within a fold. The grid uses kilometer-scale tangent-plane coordinates at latitude 49 degrees. Exact grid boundaries are arbitrary, so grouping is a dependence stress test rather than a claim that cells are biologically independent.

The selection rule is fixed before the comparison: choose the nonconstant candidate with the largest pooled development average precision. Average precision weights precision at each newly attained recall level [6]. This measures ranking in a rare-positive task. Brier score and log loss separately examine probabilities. There is no probability recalibration or threshold fitting on the later labels.

The combined model has development AP 0.6061, compared with 0.1944 for the geographic smoother and 0.0844 for text. Its Brier score is 0.01247, below the baseline 0.02040. It is selected before evaluation on the 1697 adjudicated July–October reports, which contain only six positives. Table 2 retains every candidate's result so that selection failure remains visible.

The selected model's AP falls substantially from development to the temporal holdout. This is evidence against assuming stable performance. Its mean probability is 1.595%, while

Table 2. Fixed candidates on the same later-report holdout

Candidate	AP	Brier	Log loss	Hits in 85
Prevalence	0.0035	0.003863	0.03563	0
Geographic	0.1343	0.003466	0.02293	6
Text	0.0313	0.003807	0.03419	2
Combined (selected)	0.1142	0.004701	0.02248	6

the holdout's observed positive fraction is 0.354%. Its Brier score is worse than the prevalence baseline, despite a better log loss and useful ranking. Accordingly, the fitted numbers are conditional model probabilities requiring calibration surveillance; they are not reliable real-world confidence levels.

The metrics are defined for all n held-out adjudicated reports:

$$\text{Brier} = n^{-1} \sum_i (p_i - Y_i)^2, \qquad \text{LL} = -n^{-1} \sum_i [Y_i \log p_i + (1 - Y_i) \log(1 - p_i)]. \tag{7}$$

For an investigation capacity B , precision is the number of positive reports in the highest-ranked B divided by B , and recall divides that number by all held-out positives. An all-negative classifier would appear 99.65% accurate here while finding no positive. Accuracy is therefore not our decision criterion.

4.3 Intervals, selection and dependence sensitivity

The selected model puts six of six holdout positives in its top 85 reports, the smallest integer capacity covering 5% of the holdout. Precision is 7.06%, with a conditional 95% Jeffreys interval of 3.00–13.97%; recall is 100%, with a corresponding interval of 66.96–99.99%. These binomial intervals assume exchangeable report outcomes and understate uncertainty when one episode generates repeated reports. They are not population-level coverage guarantees.

We additionally resample 10-km holdout cells 2000 times. All 2000 resamples have a defined Brier score; the full-resampling percentile interval is 0.00164–0.00954. Of these resamples, 274 (13.7%) contain no positive reports. Their AP and recall are undefined and are retained as JSON null, not replaced by zero. The AP interval 0.0175–0.3557 and recall interval 1–1 explicitly condition on the remaining 1726 positive-containing resamples. The degenerate conditional recall interval arises because the few observed positive localities were found; it supplies no evidence of perfect recall in a new outbreak. An entirely new location can still fail. The complete replicate-level export preserves Brier for every draw, along with the positive count and undefined-metric flags.

Removing the four explicit registry-style positive records and refitting without them reduces holdout positivity to four reports among 1695. In this stress test, 85 investigations find three of four positives, AP falls to 0.0413 and Brier increases to 0.00981. One can therefore not claim that the strong aggregate ranking has been established for ordinary public reports. The geographic 10- and 60-km cases both find six positives in the top 43 records, but this sensitivity result cannot retroactively replace the originally selected model or become a new independent test.

The final model is refitted on all 2083 adjudicated reports for a current queue. To expose fitting instability, spatial-cell Poisson weights are used in 160 bootstrap fits, excluding replicates with no weighted positives. Dictionary and scaling are held fixed for this calculation; its percentile intervals condition on those representations and do not include vocabulary-selection or verification-transport uncertainty. Table 3 reports the numeric coefficients and their conditional intervals; the complete fit is retained with the computational outputs. Their interpretation is predictive, penalized and conditional; they are not causal biological effects.

Table 3. Predictive numeric coefficients in the final refit

Standardized feature	Coefficient	Conditional interval
Latitude	2.723	1.584–3.317
Longitude	-0.705	-1.015—0.292
Month sine	0.172	-0.169–0.458
Month cosine	0.613	-0.321–0.930
Logarithmic lag	0.866	0.645–1.249
Invalid detection date	-0.314	-0.467–0.002
Blank Notes	0.287	-0.328–0.973

Three probability targets must be distinguished:

$$p_V(x) = P(Y = 1 \mid x, V = 1), \quad p_{\text{All}}(x) = P(Y = 1 \mid x), \quad p_U(x) = P(Y = 1 \mid x, V = 0). \quad (8)$$

The fitted score targets adjudicated reports, whereas the investigation queue contains reports already known to be unresolved. Let $a(x) = P(V = 1 \mid Y = 1, x)$ and $b(x) = P(V = 1 \mid Y = 0, x)$, with $0 < a, b < 1$. The selection-rate ratio $\rho = a/b$ gives

$$\frac{p_V}{1 - p_V} = \rho \frac{p_{\text{All}}}{1 - p_{\text{All}}}, \qquad p_{\text{All}} = \frac{p_V}{\rho(1 - p_V) + p_V}. \quad (9)$$

This inverse-selection formula recovers the all-report target, not the unresolved-report target. For example, $p_{\text{All}} = 1/10$, $a = 4/5$ and $b = 1/5$ produce $p_V = 4/13$, $\rho = 4$ and $p_U = 1/37$. Treating the inverse- ρ value $1/10$ as p_U would be wrong.

For the unresolved queue, define a different transport parameter,

$$\eta(x) = \frac{a(x)[1 - b(x)]}{b(x)[1 - a(x)]} = \frac{p_V/(1 - p_V)}{p_U/(1 - p_U)}, \qquad p_U = \frac{p_V}{\eta(1 - p_V) + p_V}. \quad (10)$$

The factors $1 - a$ and $1 - b$ account for being unresolved. Knowing ρ does not determine them, so it does not identify p_U . The supplied data do not estimate a , b or η .

Table 4. Selection-ratio counterexample

Type	p_V	a	b	ρ	p_{All}	p_U
A	4/5	4/5	1/5	4	1/2	1/5
B	7/10	1/5	1/20	4	7/19	28/85

Table 4 uses legal selection probabilities and no new pest observations. Despite the same $\rho = 4$, $p_{V,A} > p_{V,B}$ but $p_{U,A} < p_{U,B}$. The corresponding transport ratios are $\eta_A = 16$ and

$\eta_B = 19/4$. This disproves an unconditional ranking claim based on constant ρ . As a positive control, when $a = b = 1/2$, $\rho = \eta = 1$ and all three targets coincide.

The queue CSV now labels the raw fitted score as `pV_positive`, its complement as `pV_mistaken`, and unresolved sensitivity values as `pU_positive_eta_0.25`, `pU_positive_eta_1.0` and `pU_positive_eta_4.0`. These η values are declared transport scenarios, not estimates or confidence limits. A constant positive η preserves the ordering of the transformed p_U values. Feature-dependent η can reverse that ordering; constant ρ alone cannot protect it. The default queue remains a provisional ranking by p_V , requiring empirical verification and exploration rather than automatic decisions. In particular, $\eta = 1$ explicitly assumes equal positive odds in verified and unresolved reports, in addition to the still-unverified calibration of the score.

5 Allocating limited investigations

5.1 Expected yield and a protected exploration budget

For the unresolved queue, write $p_i = p_U(x_i)$ for its correct target probability. With equal-cost investigations, the expected number of confirmations from a selected set S is

$$U(S) = \sum_{i \in S} p_i, \qquad |S| \le B. \tag{11}$$

Linearity of expectation makes this expression valid even if outcomes are dependent. Under this additive objective and fixed probabilities, selecting the highest B probabilities is optimal: replacing a selected lower-probability item with an unselected higher-probability item cannot reduce U . It does not maximize the number of independent nests found, establish true probabilities, or solve variable-cost travel scheduling. Applying this theorem to the raw p_V queue additionally requires a constant verified-to-unresolved transport ratio η and adequate score calibration. Those conditions are not established here, so the supplied ranking is an investigation aid rather than a proved optimum for true unresolved-positive yield.

For unequal costs c_i and missed-positive loss L_i , a single report justifies investigation when $p_i L_i > c_i$, under the stated loss model. With a hard joint budget, maximizing $\sum_i p_i L_i z_i$ subject to $\sum_i c_i z_i \le C$ is a 0–1 knapsack problem; sorting p_i/c_i alone does not generally solve it. The supplied data include neither costs nor agency loss values, so a universal numeric threshold would be invented.

We propose reserving 20% of capacity for exploration and missing-information follow-up, with the rest following the ranking. This is a stated policy safeguard, not an empirically optimized fraction. The exploration share should include geographically novel reports, unusual credible morphological descriptions, and randomly selected reports from underexamined cells. Their adjudications must record selection probabilities so that verification bias can eventually be estimated. The public should never be told that a low score means the insect is absent.

Figure 3 shows the observed budget curve for the protected holdout. Six positives occur in the top 85 combined-model records; the text-only curve is much weaker. Under random selection of 85 from 1697 reports, expected positive yield is $85(6/1697) \approx 0.30$. The comparison

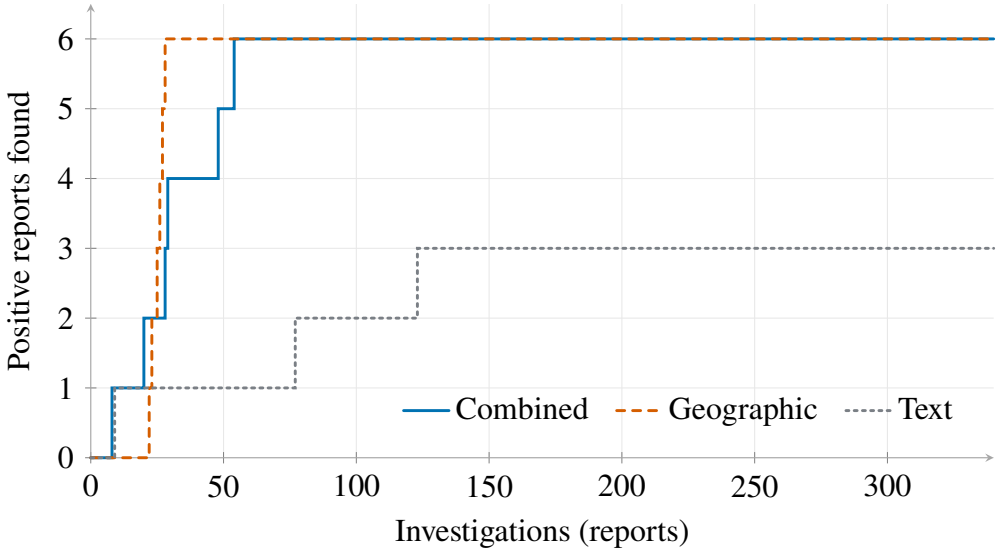


Figure 3. Investigation yield on later adjudicated reports. Each step counts a withheld positive at its frozen-model rank; no unresolved report supplies an evaluation label. The curve values are retrospective and conditional on this adjudicated sample.

is against equal-cost random sampling of adjudicated reports, not a claim that this many true positives will be found in the current unresolved queue.

5.2 A usable queue with jurisdiction and uncertainty checks

The final refit ranks 2357 unresolved reports. The complete CSV distinguishes p_V scores and complements, p_V fitting intervals for the leading 30, and p_U sensitivity scenarios under declared verified-to-unresolved transport ratios. The default rank remains by p_V ; the $\eta = 1$ scenario is an assumption of equal conditional odds, not an estimated property of unresolved reports. Ten unresolved records are north of latitude 49 degrees, outside Washington's northern land boundary. These should be routed to the appropriate Canadian contact rather than consume a Washington field visit. Remaining membership in Washington is not certified by a rectangular latitude filter: the southern boundary and address accuracy still require agency review.

Table 5. Leading unresolved candidates for jurisdiction review

ShortID	Latitude	Longitude	p_V score	p_V fitting interval
EB4746A1	48.9862	-122.6975	0.871	0.118–0.957
0CAC26D0	48.7863	-122.5506	0.808	0.091–0.936
0D889239	48.9841	-122.7744	0.768	0.218–0.911
06780761	48.7600	-122.9133	0.725	0.139–0.894
7D0FB65F	48.9241	-122.2551	0.720	0.308–0.931

Table 5 lists high-ranked candidates at or south of latitude 49 degrees; the shortID is the first eight hex characters of the unique GlobalID. These are requests for evidence, not new confirmed sightings. Follow-up should ask for a specimen or a clear existing image, precise location, detection date and whether the event is still active. The score already uses the submitted prose;

repeating it as "independent" support would double-count information.

The sum of p_V scores for the top 100 unresolved records is 21.63. This is an expected unresolved-positive yield only if the scores are calibrated and $\eta = 1$ is valid. Otherwise, it is a score aggregate. The observed temporal overprediction, regional dependence and large selection uncertainty make it unsuitable as a staffing promise. We retain it as an audit of model output, not a forecast of 21 actual hornets.

The five near-coincident late positives can represent a common event. A sensitivity fit weights each 2-km positive component as one total positive. Leading queue probabilities change markedly. A practical dispatch policy should first connect reports to existing investigations, avoid duplicate field visits, then rank distinct actionable episodes; report-level expected yield alone cannot value the first evidence from a new area.

6 Updating the model as reports arrive

Incoming reports should be scored immediately with the currently frozen model. Receipt-time features and the model version must be logged; corrections create a new record version without erasing the original. Missing or impossible dates trigger clarification, not an inferred negative. New image or specimen evidence can be referred to entomologists even though the present model uses no images.

Each week, reconcile laboratory determinations, pending reports, repeated incidents, geographic referrals and the ratio of adjudicated to unresolved reports. Separate the report arrival, sighting, coordinate conversion and label completion timestamps. Compute Brier/log loss and budget yield only on outcomes that have actually become available, and compare each score with a prevalence baseline using the same eligible reports. Keep high-ranked and exploratory investigations identifiable in the audit data.

Refitting every time a single report arrives would chase noise in a 14-positive sample. We propose a monthly review, with an immediate scientific review after a confirmed report in a previously unrepresented area or a changed reporting mechanism. If fewer than five new independent positive locality clusters have been adjudicated, retain the existing fitted coefficients and update the queue; five is an operational trigger to prevent overreaction, not a statistically established sufficiency threshold. When new evidence is adequate, refit preprocessing and model together and evaluate against a subsequently arriving block before replacing the deployed version.

A future intercept update can address a changed base rate if the adjudication mechanism is stable. It cannot repair a change in the verified-to-unresolved transport ratio $\eta(x)$, a new geographic introduction, or image-driven changes in selection. A model replacement should therefore be accepted only when ranking, calibration, missed-positive patterns and exploration coverage jointly support it. If calibration remains poor, distribute ordered priority bands rather than numerical confidence to field staff.

7 What would count as eradication evidence?

7.1 A negative result needs a detection model

Let H be the hypothesis that at least one reproducing population persists in Washington after the relevant control action. Let $s_t = P(D_t = 1 \mid H)$ be the system's probability of a confirmed detection during an active season, accounting for where the population could be and whether that location is surveyed. Under a specified conditional independence model for seasons,

$$P(D_1 = \cdots = D_K = 0 \mid H) = \prod_{t=1}^{K}(1 - s_t). \tag{12}$$

For constant sensitivity s , K zero-detection seasons have likelihood $(1 - s)^K$. Thus a 5% nondetection likelihood requires

$$K \geq \left\lceil \frac{\log(0.05)}{\log(1 - s)} \right\rceil. \tag{13}$$

Table 6 displays sensitivity cases. Each threshold is independently checked at K and $K - 1$.

Table 6. Zero-detection seasons under declared system sensitivity

Sensitivity s	Required seasons K	$(1 - s)^K$
0.10	29	0.0471
0.30	9	0.0404
0.50	5	0.0313
0.80	2	0.0400

These are hypothetical system sensitivities, not estimates from the supplied workbook. In particular, 2069 negative species identifications are not 2069 independent searches of potentially occupied nests. Public reporting has no measured spatial coverage or standardized effort, and inaccessible areas can have near-zero sensitivity. A model trained to recognize mistaken reports does not supply the sensitivity of a statewide search.

A Bayesian statement is also conditional. If prior odds of persistence are O_0 and eradication guarantees no positive detection, posterior odds after no detection are

$$O_K = O_0 \prod_{t}(1 - s_t). \tag{14}$$

With equal prior odds and $s = 0.8$, two zero-detection seasons give posterior persistence $0.04/1.04 = 3.85\%$. This is not a data-derived 95% eradication probability: both the prior and sensitivity are assumptions. Winter inactivity, dependent survey failures and reintroduction can invalidate the simple update.

7.2 An evidence standard for WSDA

A defensible eradication assessment would need all of the following: adjudication of credible active reports, designed surveillance over the plausible range, repeated visits during active

seasons, an audited lower bound on detection sensitivity, and explicit treatment of unsurveyed regions and reinvasion. The nest and queen scenarios should define where effort is needed, while randomized or otherwise recorded exploration prevents the classifier from surveying only locations it already expects to be positive.

If lower bounds $\underline{s}_t$ are valid for every plausible surviving-population configuration, then $\prod_t (1-\underline{s}_t)$ conservatively bounds nondetection likelihood. A geographically averaged sensitivity cannot certify a neglected low-detection refuge. Repeat-visit occupancy methods formalize this distinction between presence and detection [7]; the present data lack their required visit histories. New standardized surveillance data would be a future model update, not an extra input silently added to this analysis.

The supplied data already contain confirmed positives through autumn 2020. They do not document a complete eradication action followed by an adequate negative surveillance period. Therefore, they do not establish eradication in Washington. The appropriate current claim is continued detection and uncertain extent, with a plan for measuring the strength of future negative evidence.

8 Validation, limitations and recommended action

Independent raw-cell counts reproduce all report-state totals. Separate formulas reproduce withheld Brier score, AP and all budget hit counts. Split-member records show no shared 10-km cell within a development fold, and later-report membership is checked against raw submission dates. A different great-circle formula reproduces the nine spatial distances. The zero-detection threshold is checked on both sides of each integer cutoff. A further validator independently reconstructs all 2357 queue probabilities from the exported vocabulary, IDF, coefficients and numeric scaling: it implements regex token counting, sublinear TF–IDF and Euclidean normalization directly, then evaluates the logistic formula without importing the model or scikit-learn transforms. This validates the deployed computation, not probability calibration. Exact synthetic Bayes controls distinguish p_V , p_{All} and p_U , including an equal- ρ ranking reversal. A separate raw-cell bootstrap audit reproduces all 2000 Brier values and the 274 undefined AP/recall cases. These checks establish limited numerical correctness; they do not establish label timing, unbiased verification, biological independence or calibrated statewide risk.

The most material weaknesses are a 14-positive sample, a mixture of public and institutional records, several repeated local positives, missing laboratory-availability timestamps, and no image content or survey-effort denominators. The temporal evaluation provides direct evidence of probability overestimation and performance shift. The registry sensitivity further weakens ordinary-public-report claims. We preserve these failures of assumptions rather than replace them with a high accuracy figure.

WSDA should use the score as a transparent investigation aid, retain exploratory surveillance, and collect the timestamps and visit denominators needed for stronger inference. Geographic scenarios can focus attention around known detections, but independent introductions and delayed reports require a wider alert channel. An eradication announcement should depend on audited active-season surveillance sensitivity and coverage, not silence in the public reporting stream.

Memorandum to WSDA

To: Washington State Department of Agriculture
From: Modeling Team
Subject: Prioritizing hornet reports and measuring eradication evidence
Data reference: Supplied reporting records through October 23, 2020

Decision requested. Use a ranked report queue to guide follow-up, reserve capacity for reports outside the learned pattern, and begin recording the surveillance information needed to support a future eradication decision.

The supplied 4440 reports contain 14 confirmed Asian giant hornet identifications, 2069 exclusions, 2342 unverified reports and 15 unprocessed reports. The 2357 unresolved outcomes must remain unknown. Their absence of confirmation is not evidence that all are mistaken, and repeated local reports need not represent distinct nests. We obtained the original image mapping but no image pixels; the present score cannot assess insect morphology.

Our model uses the submitted notes, sighting location and dates. Laboratory comments and the final status do not enter the predictors. We selected a simple regularized model using an early development set and spatial blocks. On 1697 later adjudicated reports, its top 85 candidates contained all six positives. This is a useful retrospective result, but it does not establish perfect future detection. Some positives are related and some notes describe institutional records. Removing four explicitly institutional records from fitting and evaluation reduced performance to three of four positives in the top 85.

Use priority bands, with an evidence request. The model ranks the complete unresolved queue, with identifiers, locations and uncertainty fields. High scores should trigger a request for a specimen or an existing clear photograph, an accurate location and a corrected detection date. They are not new confirmed sightings. Check whether a record already belongs to an existing investigation before dispatching a second team. Refer records outside Washington to the appropriate jurisdiction.

Keep about four out of five available follow-ups for ranked candidates and one out of five for exploration or missing-information resolution. The 20% exploration share is a proposed operating choice, not a proven optimum. Include geographically novel reports and record the sampling probabilities of randomly selected exploratory reports. A low score should never close a credible new-area alert automatically.

The model's probabilities overestimate the positive rate in the later evaluation: its average score was 1.60%, while only 0.35% were positive. These scores target adjudicated reports; positivity among unresolved reports is not identified. Their use for that queue requires a stable verified-to-unresolved odds transport relation. Therefore, field staff should receive an ordered priority band and a reason to request evidence, rather than an apparently precise infection probability. The expected-yield totals in the computational queue are conditional model outputs and must not be treated as a promise about confirmed cases.

Memorandum to WSDA: implementation and eradication standard

Local concentration supports focused surveillance, with a wider alert channel. All nine confirmed 2020 sighting locations were within 29.894 km of a confirmed 2019 location. That retrospective pattern is consistent with using a 30-km annual establishment scale to focus attention. It is not a hard dispersal boundary or a confidence region. When only the two records submitted by January 31 were available, a later-entered Vancouver Island record lay 92.633 km away; it described a September 2019 event. Delayed reporting can resemble rapid spread, and separate introductions remain possible.

Confirmed locations are sightings rather than established nest coordinates. We recommend displaying 30-, 46- and 60-km planning envelopes around known locations to show how assumptions affect coverage. The 46-km scenario allows worker sightings up to 8 km from each end of a 30-km queen-establishment movement. The wider envelope is a sensitivity case. None supplies a probability that an unvisited area is free of the pest.

Update the queue immediately; update the science when evidence changes. Score new reports with the current frozen model at receipt. Each week, reconcile laboratory outcomes, duplicates, jurisdiction referrals, corrected dates and unresolved cases. Record detection, submission, coordinate-conversion and laboratory-completion times separately. Earlier labels were not timestamped in the supplied table, so the historical forward test may be optimistic about what was known at the time.

Review the model monthly and immediately after a confirmed geographically novel report or a major change in reporting. With fewer than five newly adjudicated independent positive locality clusters, retain existing fitted coefficients and refresh the queue. This is a stability safeguard; it is not a statistical guarantee that five clusters suffice. Evaluate proposed replacements on later-arriving reports and examine calibration as well as ranking.

Eradication needs measured search sensitivity. A string of public negative identifications is not a series of independent statewide searches. A defensible absence claim requires coverage of plausible surviving locations, repeated surveys during active seasons, adjudication of credible reports, and an audited lower bound on the chance of detecting a surviving population.

If the system would detect a persistent population with probability 0.8 in each conditionally independent active season, two seasons with no detection give a nondetection likelihood of 0.04 under persistence. If sensitivity is only 0.3, nine seasons are needed to reduce that likelihood below 0.05. Those sensitivities are examples, not estimates from the supplied records. Unknown sensitivity, unsearched refuges, winter inactivity or reintroduction prevent an eradication conclusion. The current data do not establish eradication.

Recommended next step. Deploy the queue as a transparent aid to expert investigation, protect exploratory surveillance, and add standardized visit histories and label timestamps to the reporting system. These changes connect rapid triage with evidence strong enough to assess both spread and the meaning of future silence.

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Report on Use of AI

This document was prepared with an OpenAI Codex AI author and the frozen Praxis 0.7.1 tools. The AI author read the original problem and permitted inputs, developed models and validators, executed the computations, drew the figures and wrote the English paper and memorandum. No human specimen identification, independent human calculation or field observation was performed for this document.

An author-external AI review identified a probability-target mismatch and the omission of zero-positive resamples from a Brier interval. The feedback revision distinguishes verified, all-report and unresolved probabilities, corrects the full-resampling interval, and adds independent exact controls and bootstrap audits. Original sealed versions are retained; the fitted predictor, holdout and model selection were not changed. Such review is distinct from human scientific verification. NumPy, pandas, SciPy, scikit-learn, openpyxl and the Praxis workflow were used for computation. A separate validator reads raw workbook cells and exported predictions; it does not import the model. A run failed during JSON export of NumPy vocabulary indices; the failed run and its source were preserved, the exporter was repaired and a new attempt was run.

Only the supplied tabular data, original image mapping, COMAP statement and named Penn State background were used as task data. The image pixels were not obtained. Official general-method documentation and the official USGS occupancy abstract were consulted; no same-task solution, awarded paper or code was read. Model pretraining exposure is unknown. The full conversation and token usage are not available as a measured export, so this account is a factual use summary rather than a fabricated complete transcript.

This is a historical-problem rehearsal using modern AI, not a claim of compliance with an unverified 2021 AI policy or a formal contest entry. The total-page limit is applied inclusively to this disclosure.